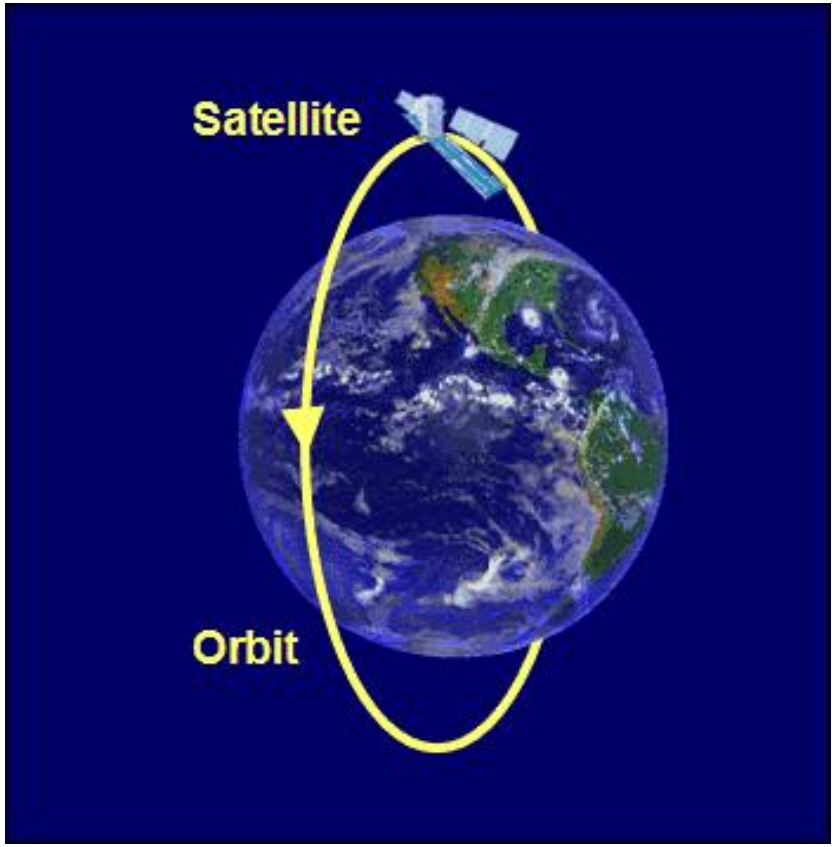
Equilibrium of Particles and Free-Body Diagrams

D. A. QUANSAH

Equilibrium of Particles

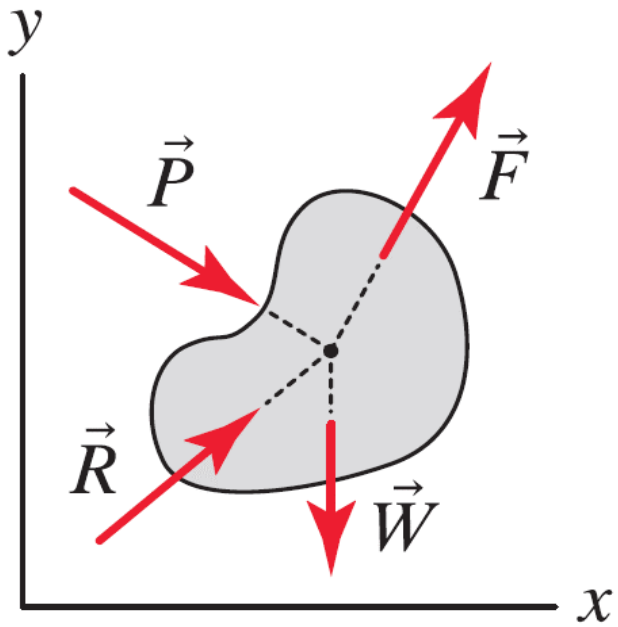
- In mechanics, a **particle** is defined to be an object whose mass is concentrated at a point and considered to have zero volume. It is also referred to as **point mass**.
- A **particle** is therefore an idealization and does not exist in reality.
- An object that is small compared to other objects and/or dimensions can often be idealized as a **particle**.
- E.g. a satellite orbiting the earth may be modelled as a particle to determine its orbit.



Equilibrium of Particles

- A large object may often be idealized as a particle, depending on the forces that are applied to the object.

- An object subjected to a concurrent force system may be idealized as a particle.

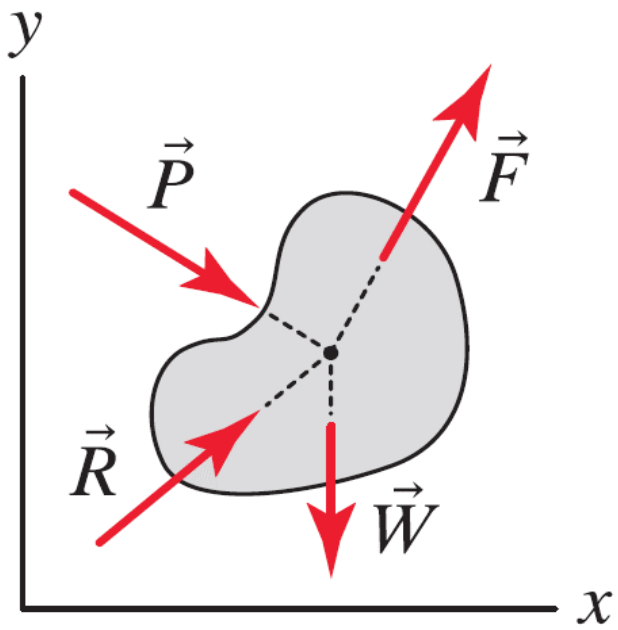


Equilibrium of Particles

- **Newton's Laws provide the conditions for static equilibrium of particles (particularly the 2nd Law).**
- **Recall:**

$$\sum \vec{F} = m\vec{a}$$

- **The summation sign is to emphasize that all forces applied to the particle must be included.**



Equilibrium of Particles

$\sum \vec{F} = m\vec{a}$

- **Static equilibrium implies (requires) that:**

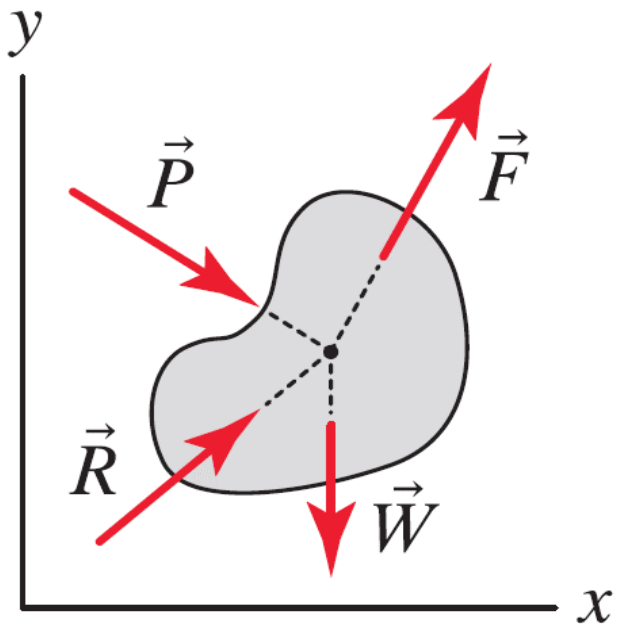
$$\vec{a} = \vec{0}$$

– **The conditions for equilibrium may be stated as:**

$$\sum \vec{F} = \vec{0}$$

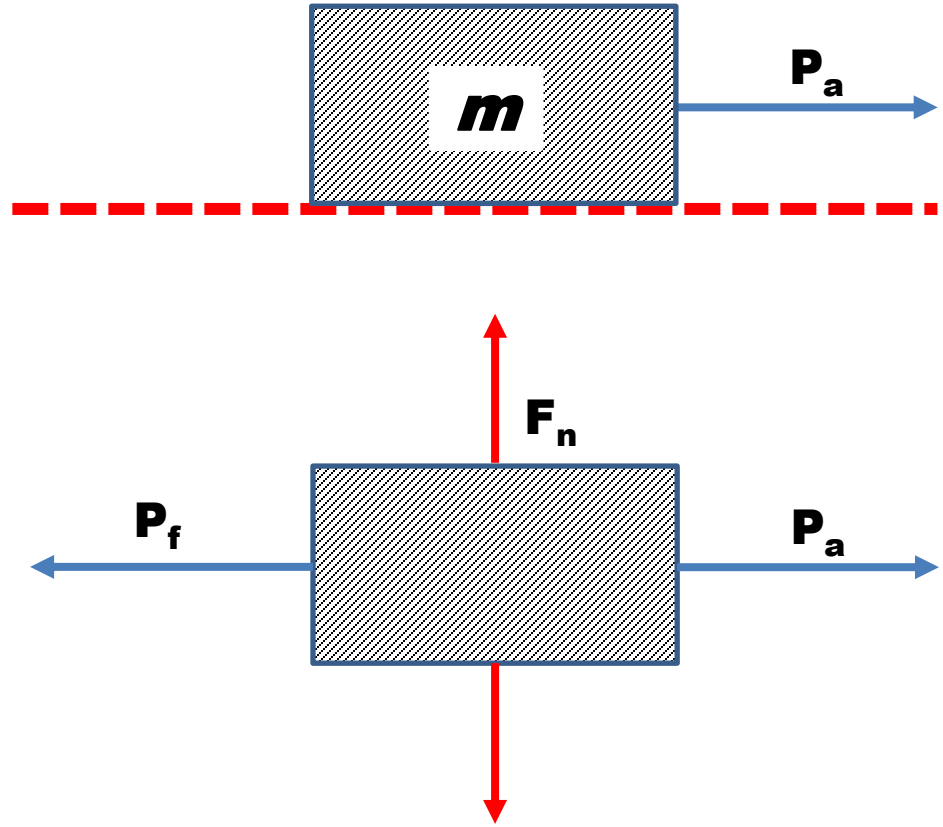
or $(\sum F_x)\hat{i} + (\sum F_y)\hat{j} = \vec{0}$

or $\sum F_x = 0$ **and** $\sum F_y = 0$



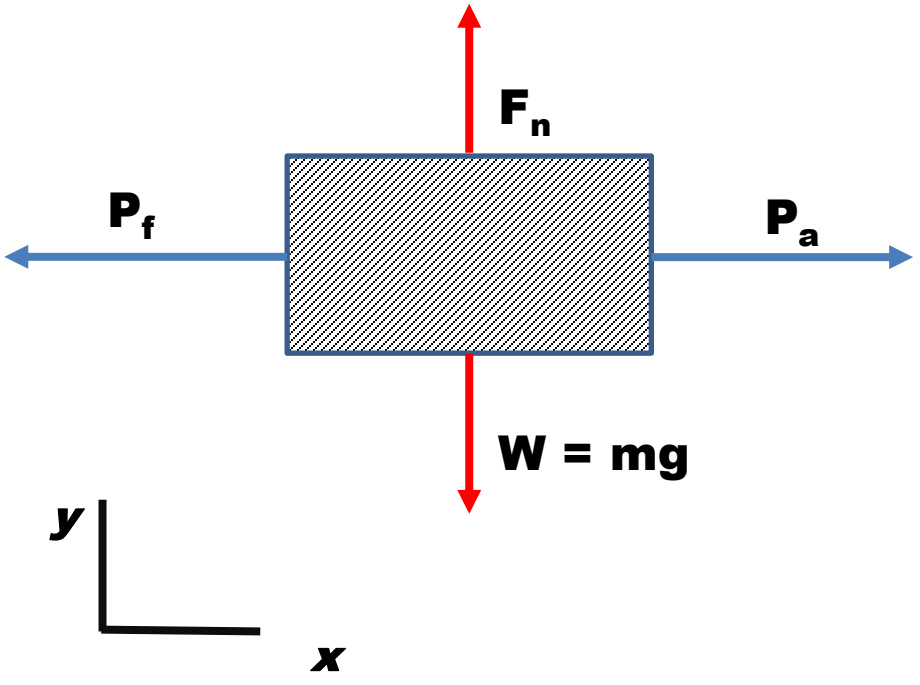
Free Body Diagrams

- **A *free body diagram* (FBD) is**
  - ✓ **a sketch of a body or a portion of a body,**
  - ✓ **that is separated or made free from its environment,**
  - ✓ **and/or other parts of the structural system,**
  - ✓ **in which *ALL* forces that act on the body are shown.**



Define a coordinate system

Free Body Diagrams



For equilibrium:

$$\sum F_x = 0, \quad \text{and} \quad \sum F_y = 0$$

Forces in x-direction:

$$P_a - P_f = 0 \qquad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

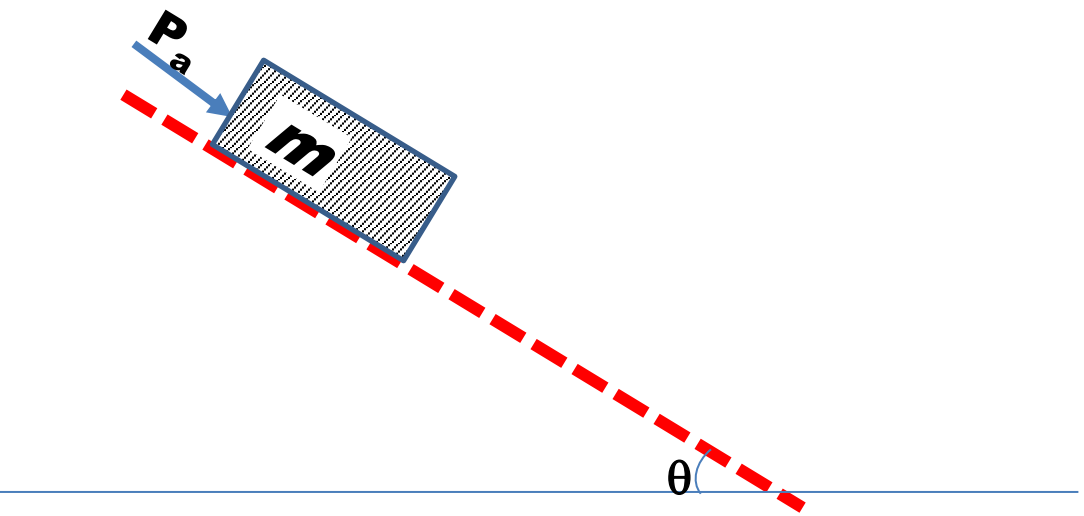
Forces in y-direction:

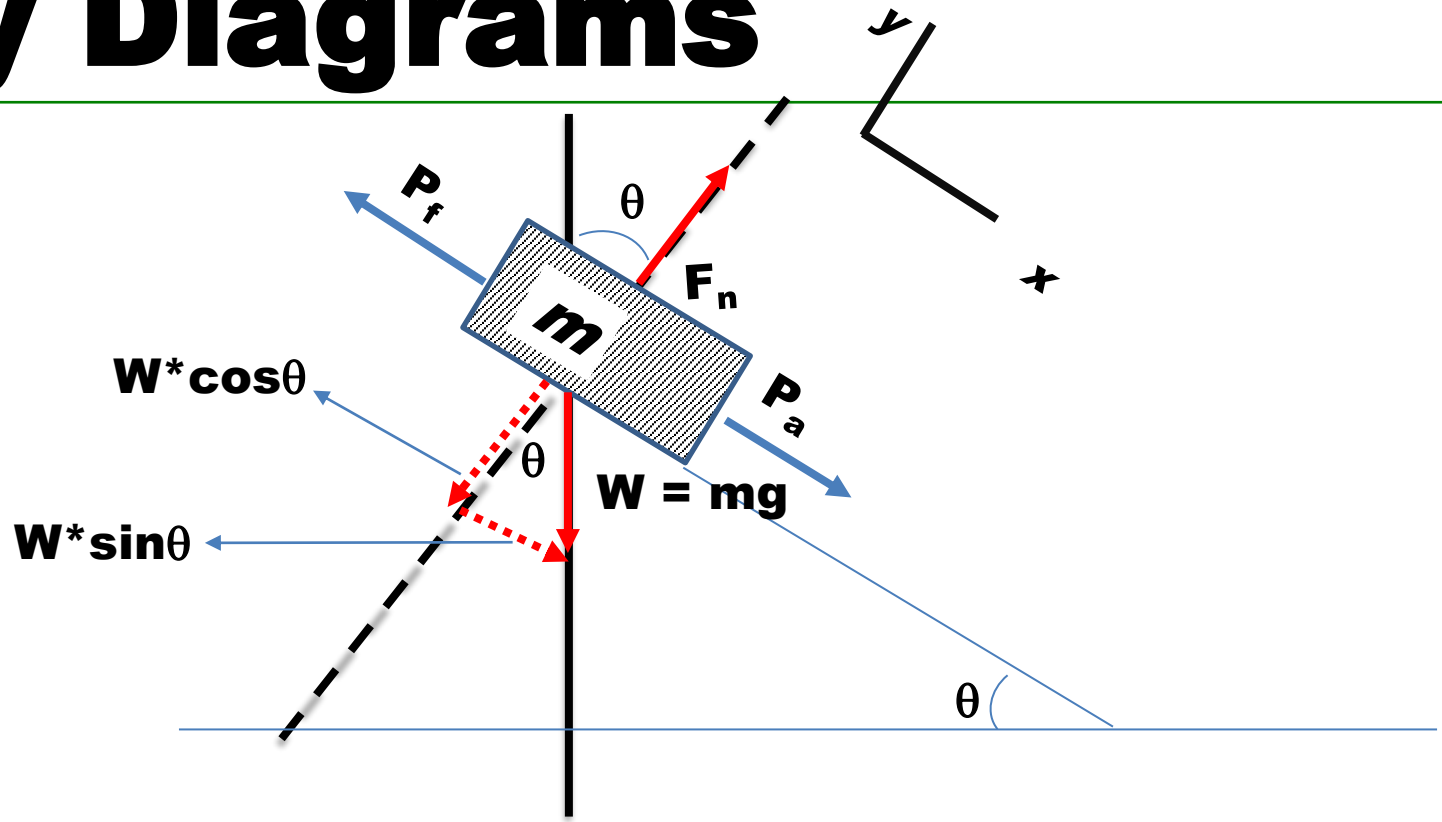
$$F_n - W = 0$$

$$F_n - mg = 0 \qquad \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

These two equations define the equilibrium condition for the system examined.

Free Body Diagrams





For equilibrium: $\sum F_x = 0$ **and** $\sum F_y = 0$

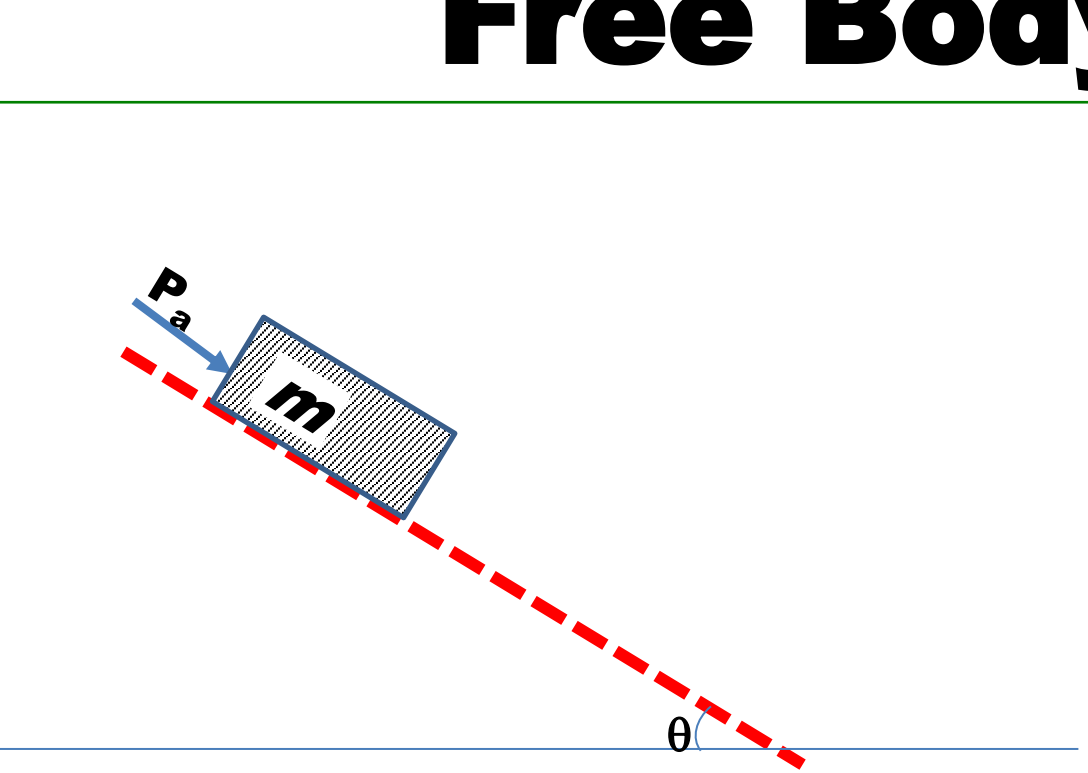
Forces in x-direction:

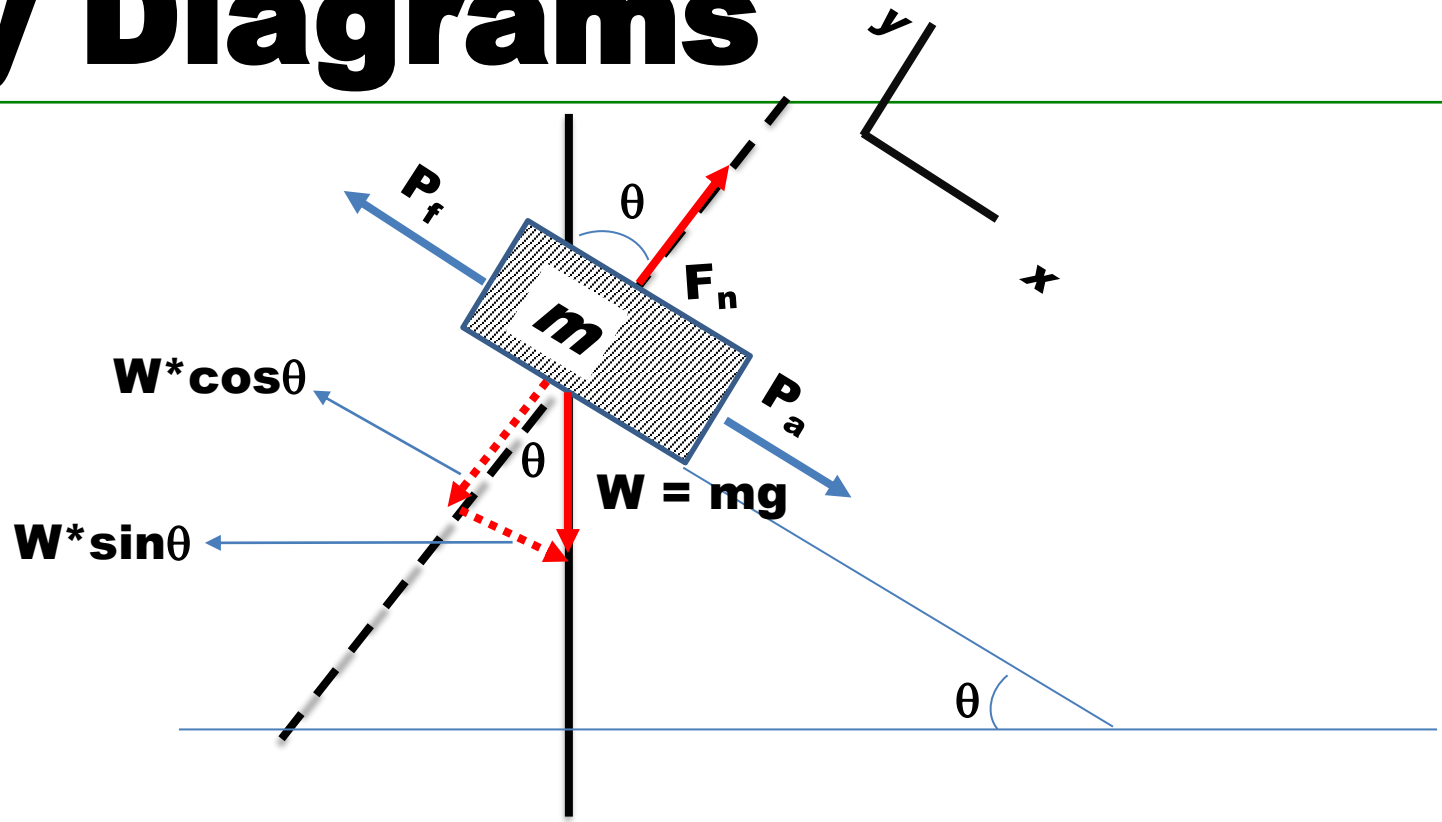
$$P_a - P_f + mg\sin\theta = 0$$

Forces in y-direction:

$$F_n - mg\cos\theta = 0$$

Free Body Diagrams





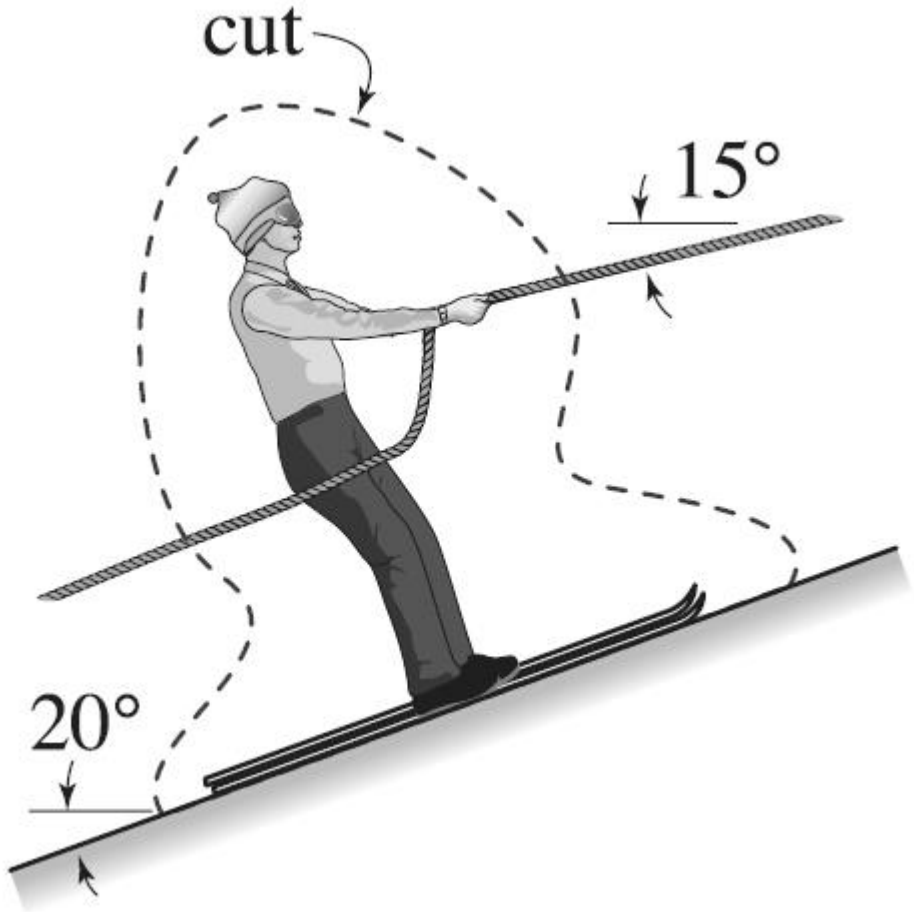
More generally (2-D cartesian):

$$P_a - P_f + mg\sin\theta = ma_x$$

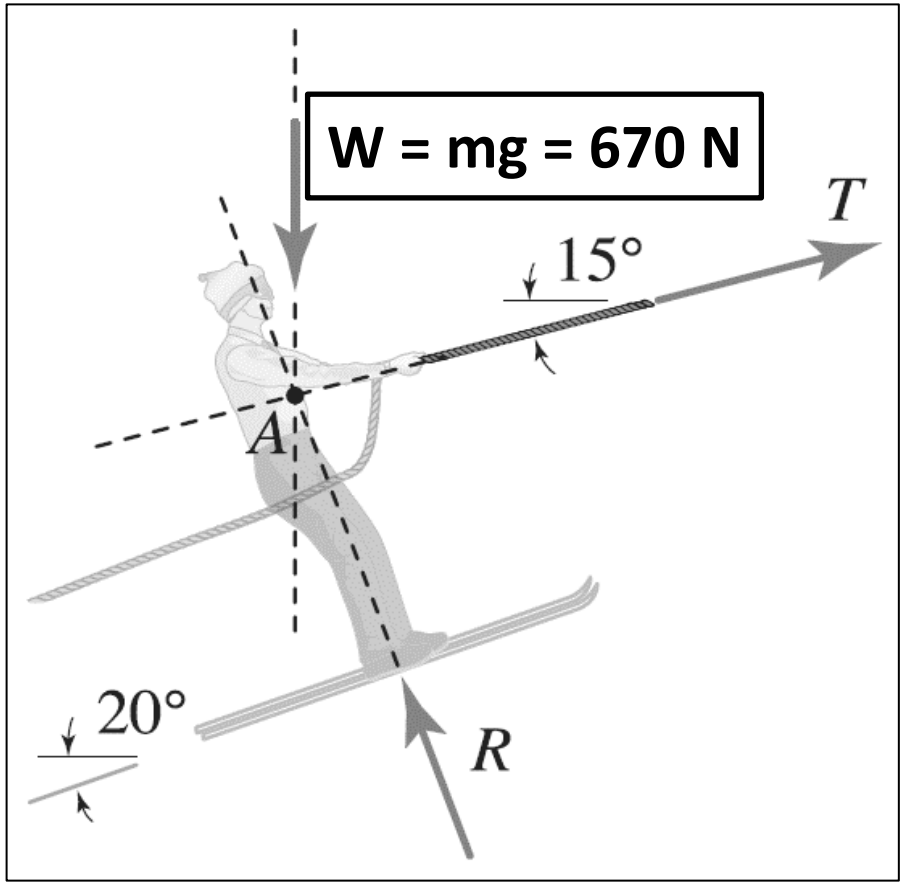
$$F_n - mg\cos\theta = ma_y$$

Free Body Diagrams

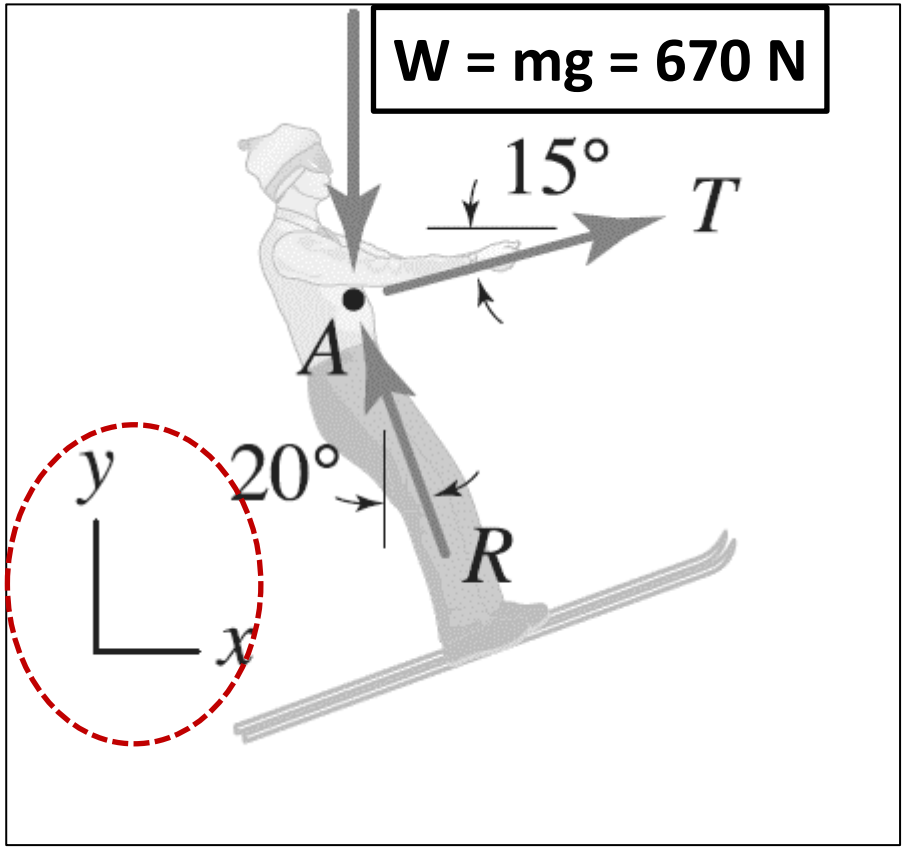
- A skier uses a tow rope as shown to reach the top of a ski hill. If the skier is of mass 68.3 kg, if the snow-covered slope can be considered frictionless, and if the portion of the tow rope behind the skier is slack, determine the force required in the tow rope to pull the skier at constant velocity.



Free Body Diagrams



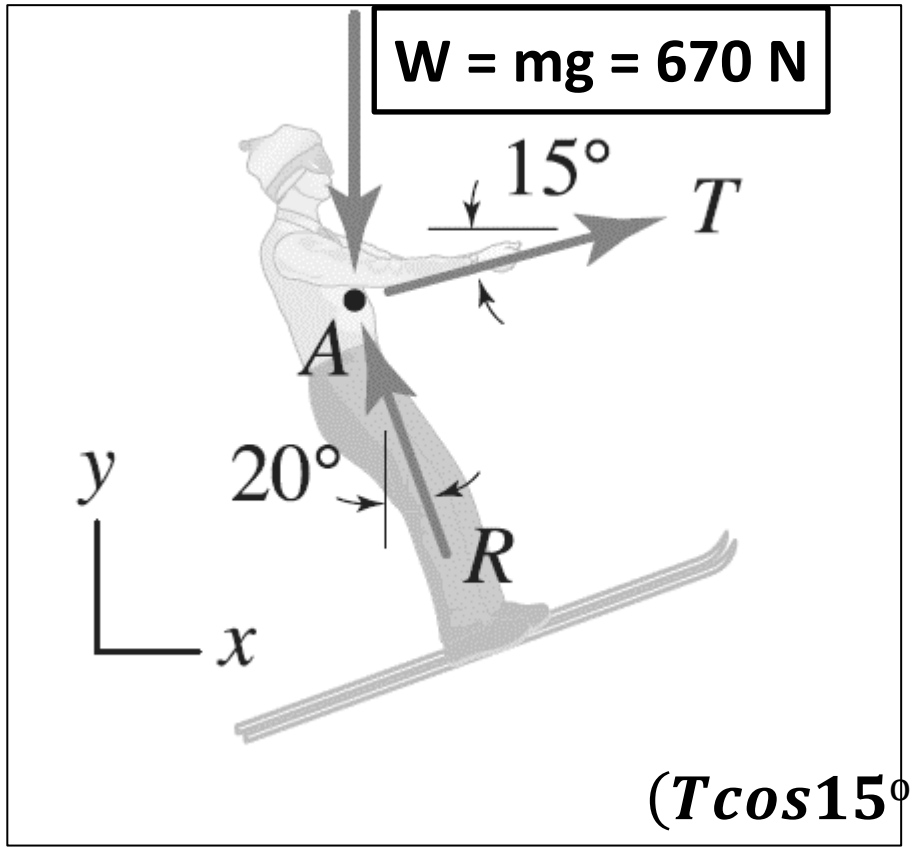
Represent all forces acting



Adopt a coordinate system

Free Body Diagrams

A vector-based approach:



$$\vec{W} = -670\text{j N}$$

$$\vec{T} = T(cos15^{o}i + sin15^{o}\text{j}) \text{ N}$$

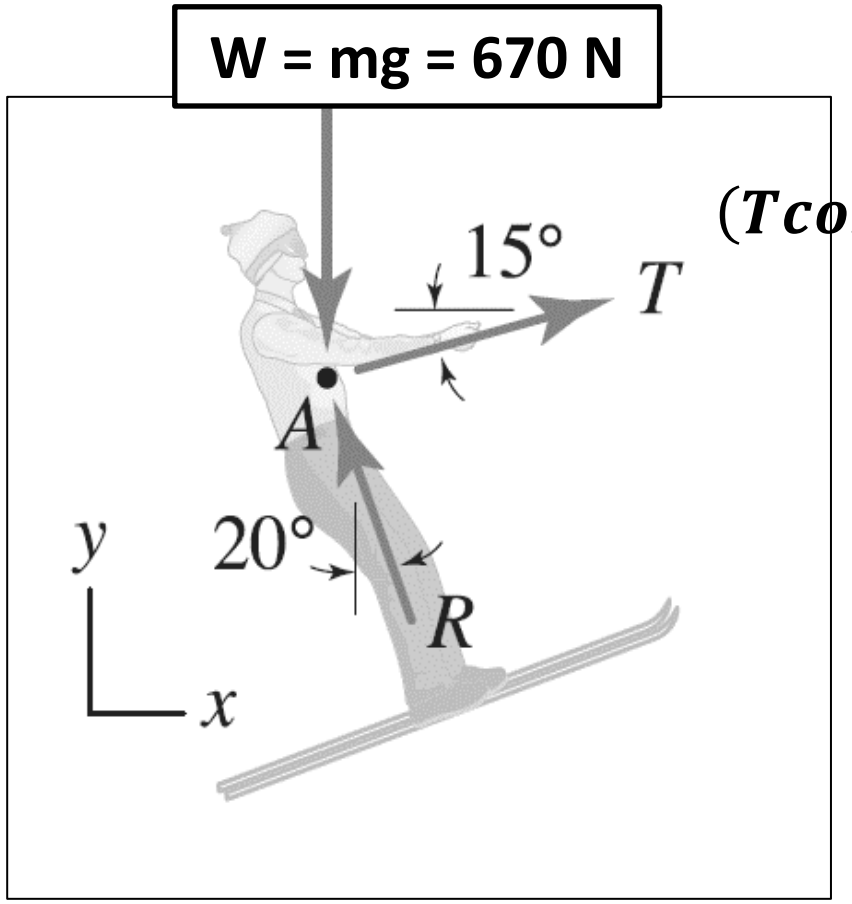
$$\vec{R} = R(-sin20^{o}i + cos20^{o}\text{j})\text{N}$$

$$\sum \vec{F} = \vec{0}$$

$$\vec{W} + \vec{T} + \vec{R} = \vec{0}$$

$$(Tcos15^{o} - Rsin20^{o})i + (-670\ N + Tsin15^{o} + Rcos20^{o})\text{j} = \vec{0}$$

Free Body Diagrams



A vector-based approach:

$$(Tcos15^{o} - Rsin20^{o})i + (-670\ N + Tsin15^{o} + Rcos20^{o})\mathrm{j} = \vec{0}$$

This requires that the scalar components of both x and y components be zero (independently)

$$Tcos15^{o} - Rsin20^{o} = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$Tsin15^{o} + Rcos20^{o} = 670 \quad \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Solving equations 1 and 2 simultaneously yields:

$$T = 230\ N \text{ and } R = 650\ N$$